

Technology Stack

Date	27 June 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

Architecture Component/ Layer	Technology Chosen	Purpose	Justification
Frontend/ Client side	React.js +Vite, HTML5, CSS3, JavaScript	Develop an interactive and responsive borrower interface.	React.js provides reusable components, Vite offers fast development, and the UI is responsive across devices.
Backend	FastAPI (Python)	Process business logic, authentication, financial analysis, and AI requests.	FastAPI delivers high-performance REST APIs with easy scalability and efficient backend processing.
Database	SQLite, SQLAlchemy ORM	Store borrower details, loan records, financial profiles,	SQLite is lightweight and reliable, while SQLAlchemy

		settlement history, and AI data.	simplifies secure database operations through ORM.
Hosting	Uvicorn ASGI Server, Python Virtual Environment (.venv), Environment Variables (.env)	Run the application securely and prepare it for deployment.	Uvicorn provides fast API execution, virtual environments isolate dependencies, and .env secures sensitive configuration values.
Version Control	Git & GitHub	Manage source code, collaboration, and project versions.	Git and GitHub enable version tracking, team collaboration, source code backup, and deployment readiness.
Third-Party APIs	Google Gemini API	AI recommendations	Generative AI